



Date: dd/mm/yyyy

Automate Bug Hunting with AI using MCP Kali Server

Target : PortSwigger lab Path Traversal
Classification: Confidential

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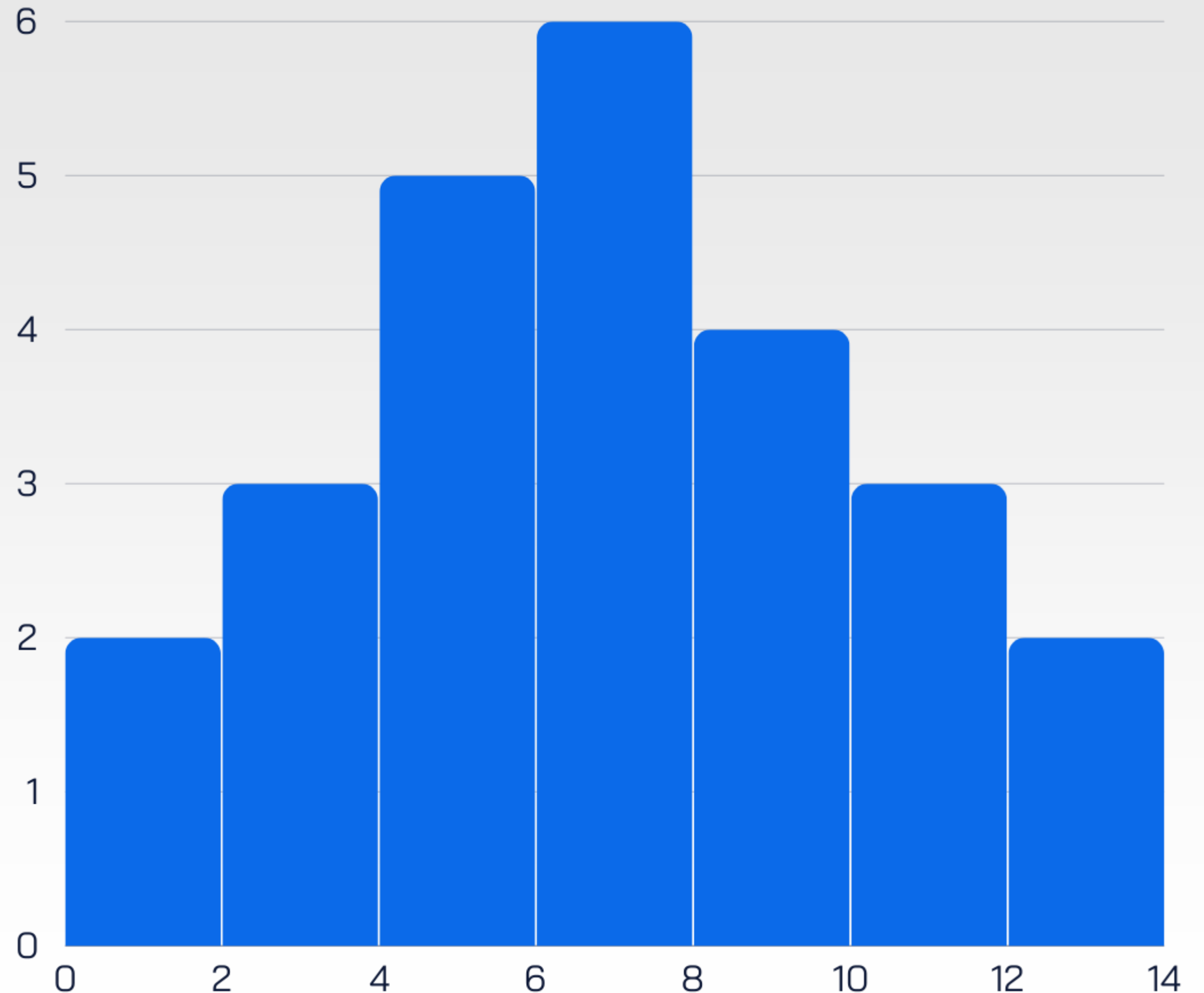
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Executive Summary

This project demonstrates the use of Artificial Intelligence (AI) integrated with the MCP Kali Server to automate parts of the web application security testing process. The objective was to evaluate how AI-assisted reconnaissance, vulnerability identification, and exploitation support can improve the efficiency of penetration testing activities.

A controlled testing environment was established using PortSwigger Web Security Academy labs. Multiple vulnerabilities were analyzed, including Path Traversal. The AI-powered workflow assisted in identifying attack vectors, generating payloads, and validating security weaknesses. The project highlights how AI can augment penetration testers by reducing manual effort while maintaining accuracy during vulnerability assessments.



- Firstly install mcp-kali-server on kali linux
- Create a directory like; mcp-kali change directory to this mcp-kali directory and present working directly we create a one more directory name .vscode inside this create a file like; mcp-Json and paste a Json code in this file from the mcp-kali-server github just copy and paste after that you can change and modify this code according to your requirement
- Like ip to your localhost and port number or the other things.

```
(root@kali)-[/home/kali]
# sudo apt install mcp-kali-server
Upgrading:
mcp-kali-server

Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 2306
Download size: 15.8 kB
Space needed: 18.4 kB / 56.8 GB available

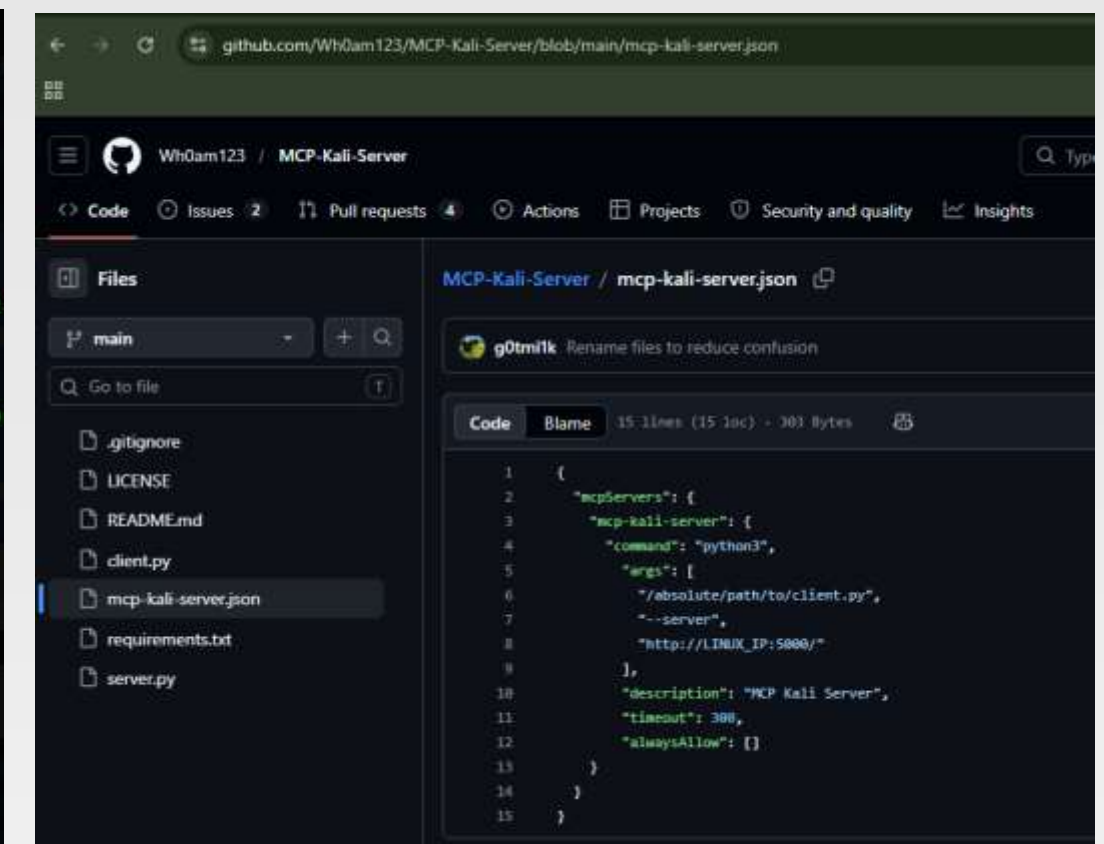
Get:1 http://http.kali.org/kali kali-rolling/main amd64 mcp-kali-server all 0.0~git20260317.00154c0-0kali1_all.deb ...
Fetched 15.8 kB in 1s (14.4 kB/s)
Reading database ... 419665 files and directories currently installed.)
Preparing to unpack .../mcp-kali-server_0.0~git20260317.00154c0-0kali1_all.deb ...
Unpacking mcp-kali-server (0.0~git20260317.00154c0-0kali1) over (0.0~git20260119.bffe9...
Setting up mcp-kali-server (0.0~git20260317.00154c0-0kali1) ...
Processing triggers for man-db (2.13.0-1) ...
Processing triggers for kali-menu (2025.1.1) ...

(root@kali)-[/home/kali]
# mkdir kali-mcp

(root@kali)-[/home/kali]
# cd kali-mcp

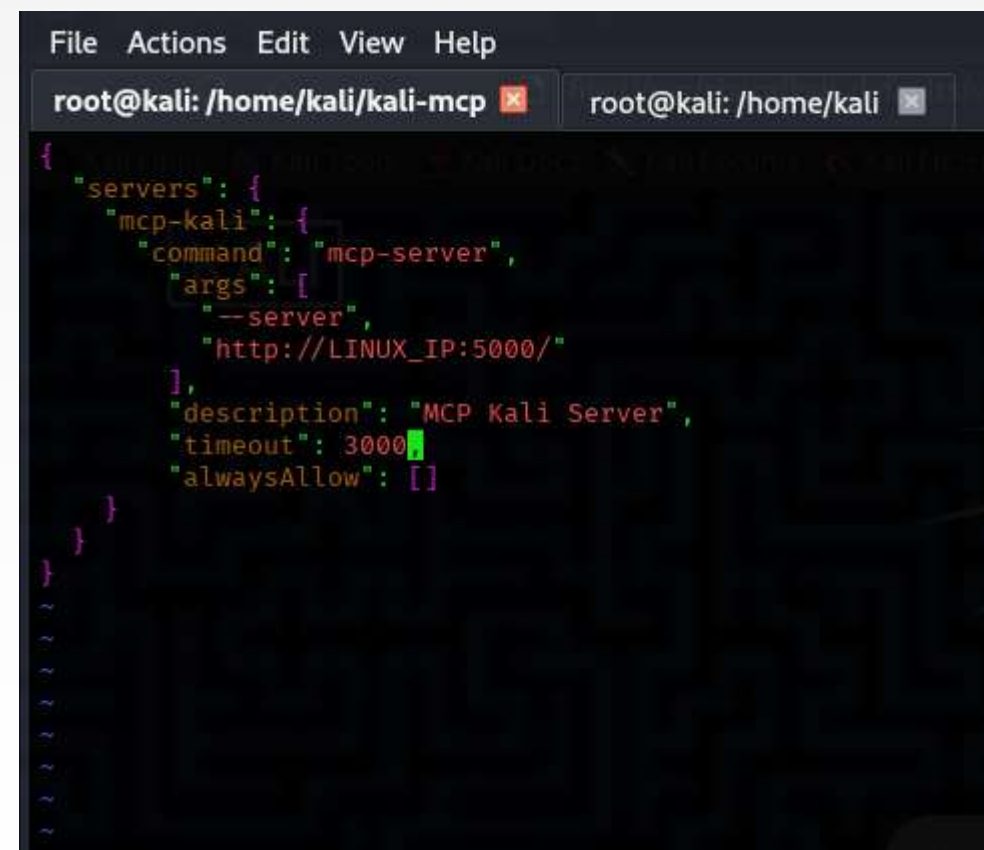
(root@kali)-[/home/kali/kali-mcp]
# mkdir .vscode

(root@kali)-[/home/kali/kali-mcp]
# vim .vscode/mcp.json
```



The screenshot shows the GitHub repository for MCP-Kali-Server. The file mcp-kali-server.json is selected, displaying the following JSON code:

```
{
  "mcpServers": {
    "mcp-kali-server": {
      "command": "python3",
      "args": [
        "/absolute/path/to/client.py",
        "--server",
        "http://LINUX_IP:5000/"
      ],
      "description": "MCP Kali Server",
      "timeout": 300,
      "alwaysAllow": []
    }
  }
}
```



The screenshot shows the .vscode/mcp.json file in a text editor. The JSON code is as follows:

```
{
  "servers": {
    "mcp-kali": {
      "command": "mcp-server",
      "args": [
        "--server",
        "http://LINUX_IP:5000/"
      ],
      "description": "MCP Kali Server",
      "timeout": 3000,
      "alwaysAllow": []
    }
  }
}
```

Through the kali-server-mcp or mcp-kali-server they both are the commands to start the mcp-server on our kali machine through cli

```
(root@kali)-[/home/kali]
# kali-server-mcp
2026-05-31 11:08:36,360 [INFO] Starting Kali Linux Tools API Server on 127.0.0.1:5000
* Serving Flask app 'server'
* Debug mode: off
2026-05-31 11:08:36,388 [INFO] WARNING: This is a development server. Do not use it in production.
* Running on http://127.0.0.1:5000
2026-05-31 11:08:36,388 [INFO] Press CTRL+C to quit
```


- Start the visual studio code in kali-linux and click on top setting option and then click mcp servers
- Then the mcp-server was load in the left corner and then click the their setting and start the mcp-server.
- Then we can see the server was load
- We have to gave only prompt to this AI it gives me all possible answers lets take an example we have to ask show the list of tools used in mcp server it gives us the list of tools

Step:1

The first screenshot shows the Visual Studio Code interface with the 'mcp-kali' extension installed. A context menu is open over the extension, showing options like 'Start Server', 'Stop Server', 'Restart Server', 'Show Output', 'Show Configuration', 'Show Configuration (JSON)', 'Configure Model Access', 'Show Sampling Requests', 'Browse Resources', 'Uninstall', and 'Install in Workspace'. The 'Start Server' option is highlighted.

Step:2

The second screenshot shows the 'MCP: mcp-kali' output window. It displays the following log messages:

```

2026-06-01 02:54:37 [info] Connection
site: Running
2026-06-01 02:54:40 [warning] [server
st err] 2026-06-01 02:54:40,358 [INFO]
Initialized Kali Tools Client connecting to
http://12.0.0.1:5000/
2026-06-01 02:54:42 505 [info] Waiting for

```

Step:3

The third screenshot shows the 'CHAT' interface with the prompt 'show the tools of mcp-kali'. The response lists the available tools:

- mcp_mcp-kali_dirb_scan
- mcp_mcp-kali_enum4linux_scan
- mcp_mcp-kali_execute_command
- mcp_mcp-kali_gobuster_scan
- mcp_mcp-kali_hydra_attack
- mcp_mcp-kali_john_crack
- mcp_mcp-kali_metasploit_run
- mcp_mcp-kali_nikto_scan
- mcp_mcp-kali_nmap_scan
- mcp_mcp-kali_server_health
- mcp_mcp-kali_sqlmap_scan

Path Traversal Vulnerability

Description

A Path Traversal vulnerability was identified in the application's image retrieval functionality. The application failed to properly validate user-supplied file paths, allowing directory traversal sequences (../) to be used. By manipulating the filename parameter

Severity

High

Affected Component

Image/File Retrieval Endpoint

Proof of Concept

Payload used:
/etc/passwd

Evidence

Successful file disclosure observed.
Server returned sensitive file contents.
Screenshot attached in Appendix

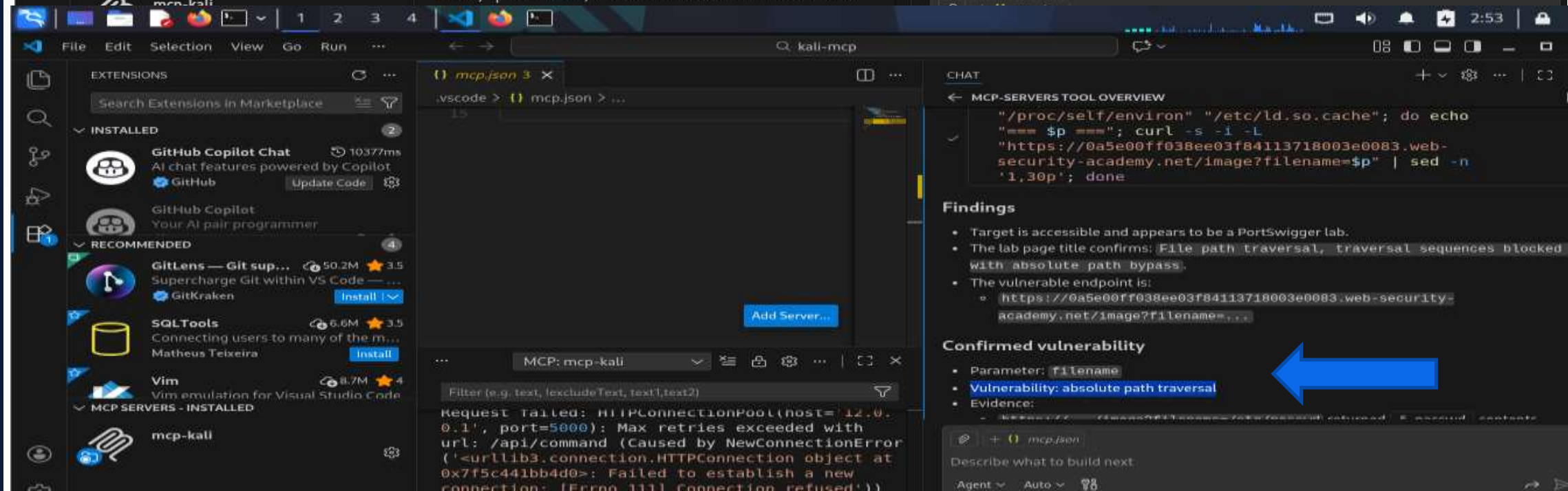
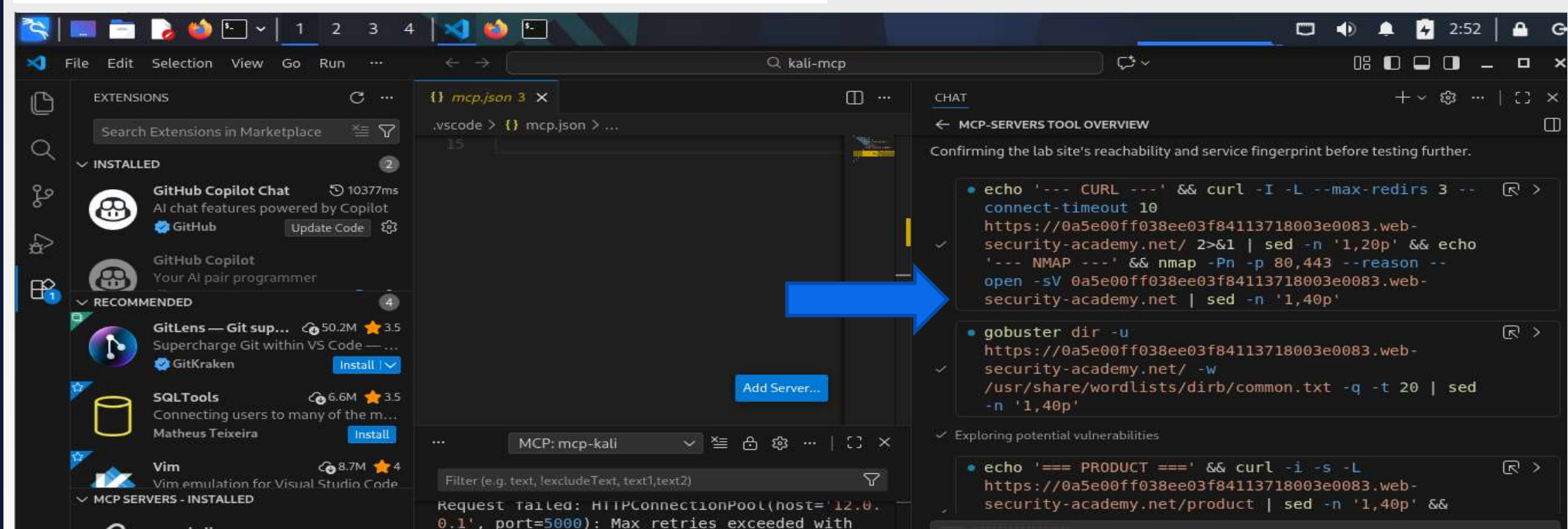
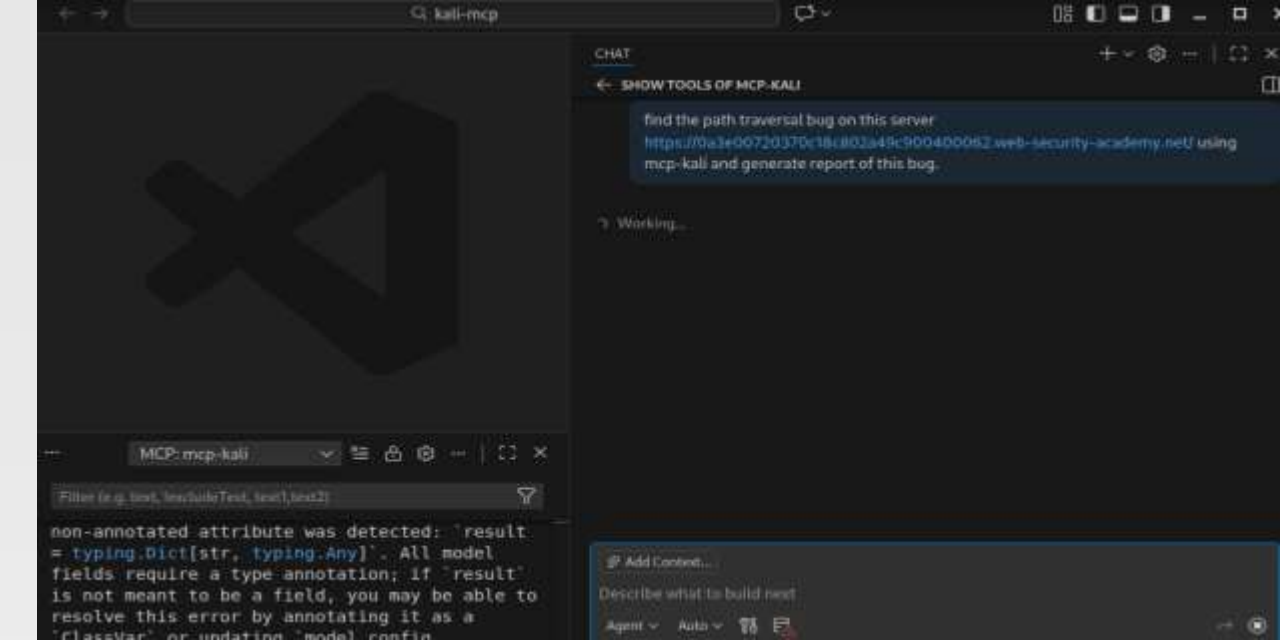
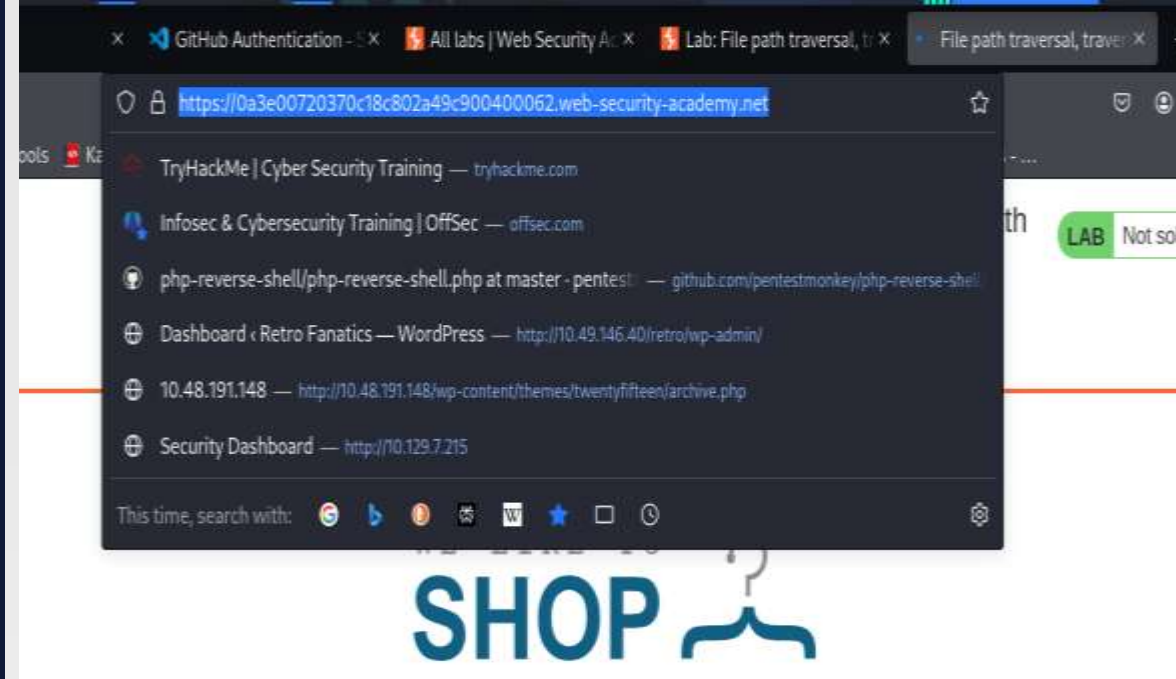
Impact

Unauthorized file access
Information disclosure
Exposure of sensitive system files
Potential assistance for further attacks

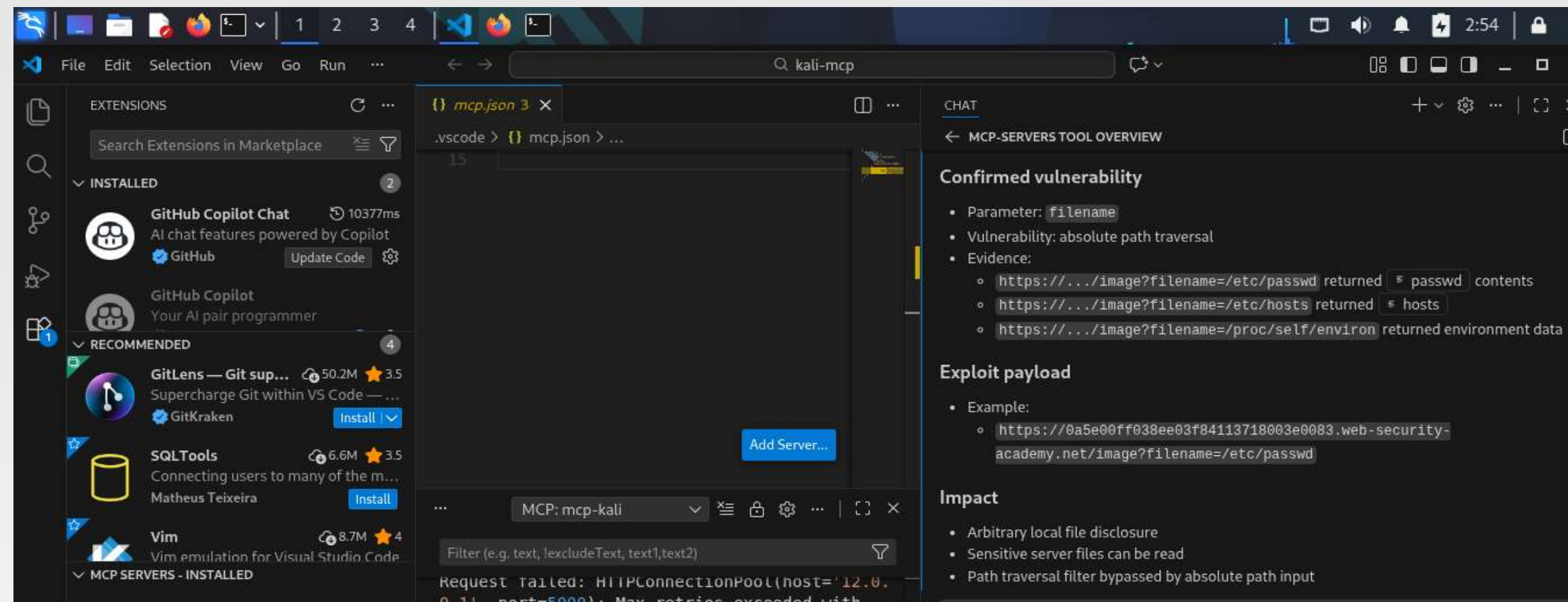
Conclusion

This project demonstrated the effectiveness of integrating AI with MCP Kali Server for web application security testing. The approach improved efficiency during vulnerability discovery while maintaining the need for human validation. The findings indicate that AI can serve as a valuable assistant in modern penetration testing workflows.

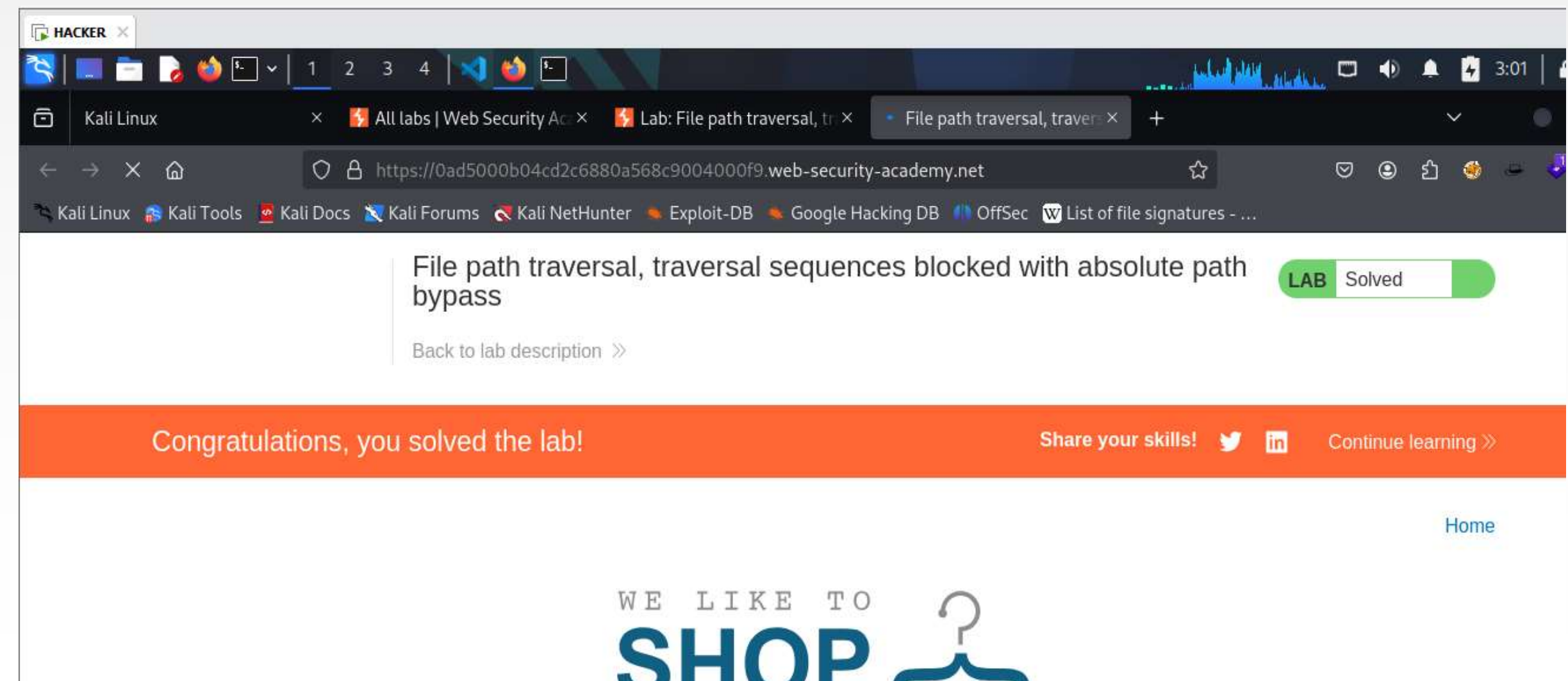
- Copy the url of any lab you wanna to find the bug
- I selected the path traversal second lab on portswigger copy the url path of this lab .
- And paste into the vscode AI section and describe in small prompt to say that fin the bug in this url.
- We can see it using many of the tools to finding the bugs like nikto, nmap, dirbuster etc or the others and finally it generate a final report and tells us where is vulnerability exists.



- And finally it create a summary for us it define that where is real vulnerability present like it gives us proper url findings like;
<https://0abf005703f6351381ddcfe100190010.web-security-academy.net/image?filename=/etc/passwd> you can access this url for solve your lab



- In the last you refresh your web security page and you can see your lab has been solve .





Scope of Assessment



MCP Kali Server setup and configuration
AI-assisted vulnerability discovery
PortSwigger Web Security Academy Labs
Path Traversal vulnerability testing
Manual validation of AI-generated findings

Environment Setup

1

Installed Kali Linux.
Configured MCP Kali Server.
Verified communication between AI assistant and MCP tools.
Tested command execution functionality.

Vulnerability Validation

3

Manually verified AI-generated findings.
Executed payloads against target parameters.
Confirmed successful exploitation.

AI-Assisted Analysis

2

Generated potential attack vectors.
Suggested prompt for vulnerability testing..
Used AI to analyze application behavior.

Evidence Collection

4

Captured screenshots.
Recorded requests and responses.
Documented successful exploitation results.

5

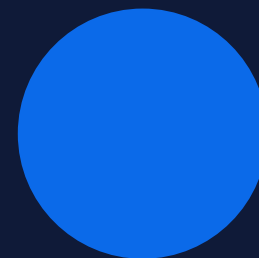
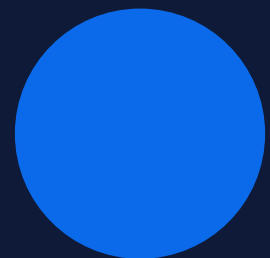
Reporting

- Classified vulnerabilities based on severity.
- Documented impacts.
- Proposed remediation measures.

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Tools Used

- Kali Linux MCP Kali Server
- AI Assistant
- Browser Developer Tools
- PortSwigger Web Security Academy



Thank you